

- ◆ $kA = k[a_{ij}]_{m \times n} = [k(a_{ij})]_{m \times n}$
- ◆ $-A = (-1)A$
- ◆ $A - B = A + (-1)B$
- ◆ $A + B = B + A$
- ◆ $(A + B) + C = A + (B + C)$, where A, B and C are of same order.
- ◆ $k(A + B) = kA + kB$, where A and B are of same order, k is constant.
- ◆ $(k + l)A = kA + lA$, where k and l are constant.
- ◆ If $A = [a_{ij}]_{m \times n}$ and $B = [b_{jk}]_{n \times p}$, then $AB = C = [c_{ik}]_{m \times p}$, where $c_{ik} = \sum_{j=1}^n a_{ij} b_{jk}$
- ◆ (i) $A(BC) = (AB)C$, (ii) $A(B + C) = AB + AC$, (iii) $(A + B)C = AC + BC$
- ◆ If $A = [a_{ij}]_{m \times n}$, then A' or $A^T = [a_{ji}]_{n \times m}$
- ◆ (i) $(A')' = A$, (ii) $(kA)' = kA'$, (iii) $(A + B)' = A' + B'$, (iv) $(AB)' = B'A'$
- ◆ A is a symmetric matrix if $A' = A$.
- ◆ A is a skew symmetric matrix if $A' = -A$.
- ◆ Any square matrix can be represented as the sum of a symmetric and a skew symmetric matrix.
- ◆ If A and B are two square matrices such that $AB = BA = I$, then B is the inverse matrix of A and is denoted by A^{-1} and A is the inverse of B.
- ◆ Inverse of a square matrix, if it exists, is unique.



NOTES

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